

James Ruse AHS Year 12 Mathematics Extension 1 Term 1 2001

- Time allowed 85 minutes.
- Attempt ALL questions.
- All questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for carelessly or badly arranged work.
- Standard integrals are printed on page 4.
- Answer each question on a new page.

Question 1

Marks

(a) Differentiate:

5

(i) $\ln(1 + e^x)$

(ii) $\ln\left(\frac{2x+1}{3x+2}\right)$

(iii) $\frac{e^{3x}}{x^2}$

(b) Find the indefinite integrals of:

5

(i) $e^{\frac{-x}{a}}$ where a is constant.

(ii) $\frac{x^3}{2-x^2}$

Question 2 Start a new page.

(a) (i) If $y = \tan 3x$ find the value of $\frac{dy}{dx}$ when $x = \frac{\pi}{3}$.

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(ii) Hence, find the equation of the tangent to the curve $y = \tan 3x$ at the point $(\frac{\pi}{3}, 0)$ (b) If $f(x) = (ax+b)\sin x + (cx+d)\cos x$, determine the values of the constants a, b, c & d such that $f'(x) = x \cos x$.

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(c) (i) Differentiate $x \tan x$ with respect to x

3

(ii) Hence find $\int x \sec^2 x dx$

Question 3 Start a new page. Marks

(a) A filter is in the shape of an inverted right circular cone of base radius 2cm and altitude 3cm. If water is flowing out of the bottom at a rate of $5\text{cm}^3/\text{min}$, find the exact rate at which level of the water is falling when the depth is 2cm.

3

(b) Prove by mathematical induction for $n \geq 1$ that:

4

$$1.2^2 + 2.3^2 + 3.4^2 + \dots + n(n+1)^2 = \frac{1}{12}n(n+1)(n+2)(3n+5)$$

(c) If $f(x) = g(x) - \ln[g(x)+1]$

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(i) Prove that $f'(x) = \frac{g(x) \cdot g'(x)}{g(x)+1}$.(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\sin x + 1} dx$

Question 4 Start a new page.

(a) Using the fact that $2 \cos^2 x = 1 + \cos 2x$, prove that $8 \cos^4 x = 3 + 4 \cos 2x + \cos 4x$.

2

(b) (i) Sketch on the same axes, the curves $y = \cos x$ and $y = \cos^2 x$, for $0 \leq x \leq \frac{\pi}{2}$.

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(ii) Find the area enclosed between these curves.

(iii) Find the volume generated when the area from (ii) is rotated about the x axis.

Question 5 Start a new page.

(a) (i) Prove that $\cot x + \tan x = 2 \operatorname{cosec} 2x$.

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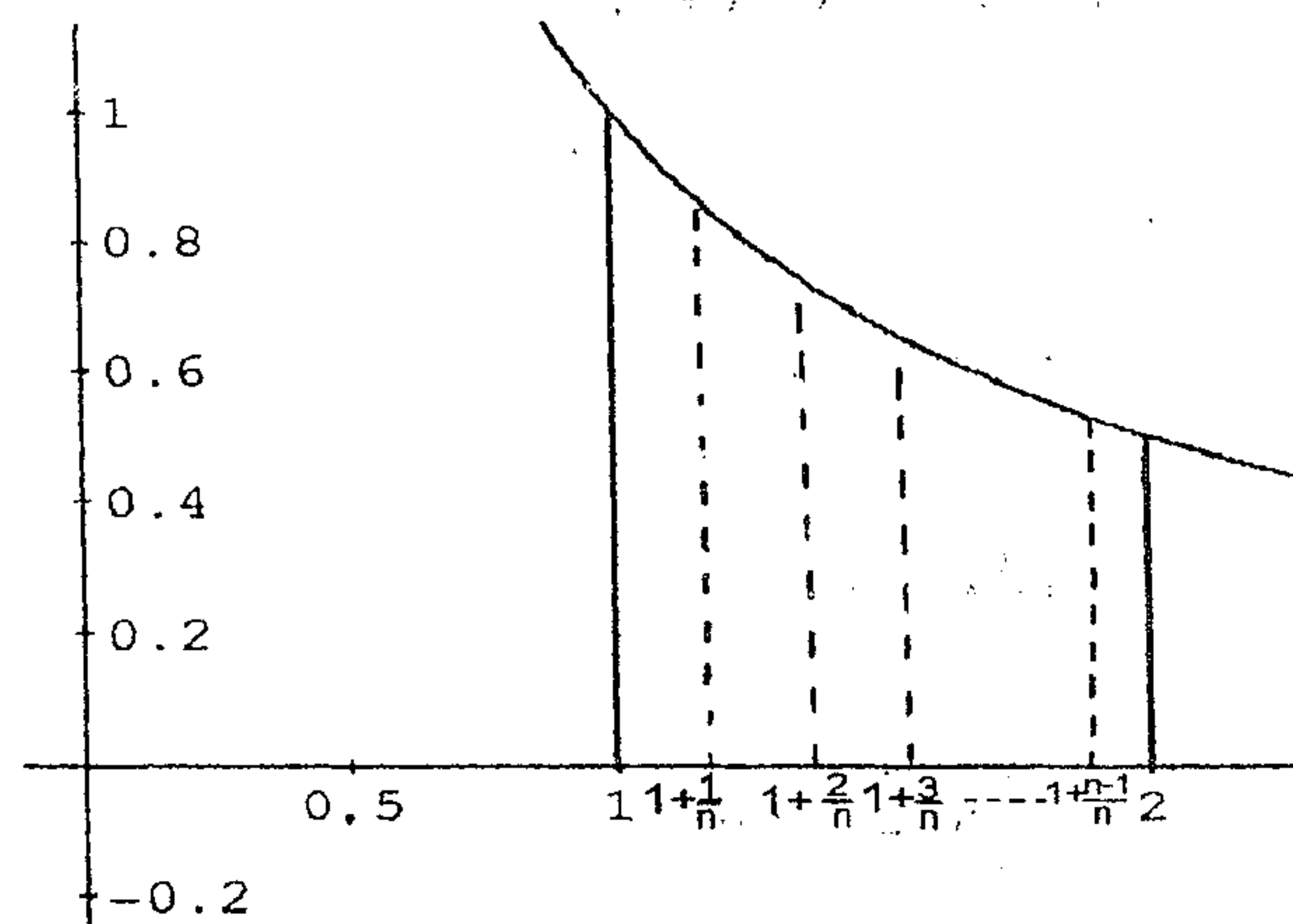
(ii) Hence evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 2 \operatorname{cosec} 2x dx$.(b) Given that $a^x = b^y = (ab)^z$, prove that $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$.

3

Question 5 (cont.)

Marks

(c)



Consider the curve $y = \frac{1}{x}$ for $x > 0$. Divide the interval from $x = 1$ to $x = 2$ into n

3

equal parts, each of width $\frac{1}{n}$. From the definition of the definite integral show that:

$$\lim_{n \rightarrow \infty} \left\{ \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right\} = \ln 2$$

Question 6 Start a new page.

(a) Determine the values of k for which $y = e^{kx}$ satisfies the equation

2

$$\frac{d^2 y}{dx^2} + 7 \frac{dy}{dx} + 12y = 0.$$

(b) A prize fund is established with a single investment of \$2000 to provide an **annual** prize of \$150. The fund accrues interest at 5% p.a. **paid half yearly**. If the first prize is awarded one year after the fund is established:

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- Find the amount in the fund account after the first prize is awarded.
- Show that the amount in the fund account after the 6th prize is awarded is approximately \$1660.
- How many prizes can be awarded before the fund is exhausted?

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

$$a(i) \frac{d}{dx} \ln(1+e^x)$$

$$= \frac{e^x}{1+e^x}$$

$$(ii) \frac{d}{dx} \ln\left(\frac{2x+1}{3x+2}\right)$$

$$= \frac{d}{dx} (\ln(2x+1) - \ln(3x+2))$$

$$= \frac{2}{2x+1} - \frac{3}{3x+2}$$

$$= \frac{1}{(2x+1)(3x+2)}$$

$$(iii) \frac{d}{dx} \frac{e^{3x}}{x^2} = \frac{x^2 \cdot 3e^{3x} - e^{3x} \cdot 2x}{x^4}$$

$$= \frac{e^{3x}(3x-2)}{x^3}$$

$$b(i) \int e^{-\frac{x}{a}} dx$$

$$= -a e^{-\frac{x}{a}} + C$$

$$(ii) \int \frac{x^3}{2-x^2} dx$$

$$= \int \left(-x + \frac{2x}{2-x^2}\right) dx$$

$$= -\frac{x^2}{2} - \ln(2-x^2) + C$$

$$2a \quad y = \tan 3x$$

$$y' = 3 \sec^2 3x$$

$$= 3 \sec^2 \frac{\pi}{3}$$

$$= 3 \text{ at } x = \frac{\pi}{3}$$

$$\therefore \text{tangent } y = 3\left(x - \frac{\pi}{3}\right)$$

$$y = 3x - \pi$$

$$2b \quad f(x) = (ax+b)\sin x + (cx+d)\cos x$$

$$f'(x) = (ax+b)\cos x + a\sin x$$

$$+ (cx+d)(-\sin x) + c\cos x$$

$$= \sin x(a-cx-d) + \cos x(ax+b+c)$$

$$\text{Now } f'(x) = x \cos x$$

$$\therefore a=1, b+c=0$$

$$c=0 \therefore b=0$$

$$a-d=0$$

$$\therefore d=1$$

$$2c(i) \frac{d}{dx} \tan x = x \sec^2 x + \tan x$$

$$(iii) \int x \sec^2 x dx = x \tan x - \int \tan x dx$$

$$= x \tan x - \ln |\cos x| + C$$

$$\frac{\pi}{3}$$

$$\frac{r}{2} = \frac{h}{3}$$

$$r = \frac{2h}{3}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \cdot \frac{4h^2}{9} \cdot h$$

$$= \frac{4\pi}{27} h^3$$

$$\frac{dV}{dh} = \frac{4\pi}{9} h^2$$

$$\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$$

$$= \frac{9}{4\pi h^2} \cdot 5$$

$$= \frac{45}{16\pi} \text{ cm/min}$$

$$1 + \frac{45}{16\pi} + \frac{45}{16\pi} \text{ cm/min}$$



Prove true for $n=1$

$$\text{LHS} = 1 \cdot 2^2$$

$$\text{HS} = \frac{1}{12} (1)(2)(3)(8)$$

$$= 4$$

$$\text{LHS} = \text{RHS}$$

Assume true for $n=k$ i.e. assume

$$1 \cdot 2 \cdot 3 \cdots k(k+1) = \frac{1}{12} k(k+1)(k+2)(3k+5)$$

4 Prove true for $n=k+1$ i.e. Prove

$$\frac{1}{12} k(k+1)(k+2)(3k+5) + (k+1)(k+2)$$

$$= \frac{1}{12} (k+1)(k+2)(k+3)(3k+8)$$

$$\text{HS} = \frac{1}{12} k(k+1)(k+2)(3k+5) + \frac{12(k+1)(k+2)}{12}$$

$$= \frac{1}{12} (k+1)(k+2) [k(3k+5) + 12(k+2)]$$

$$= \frac{1}{12} (k+1)(k+2) [3k^2 + 17k + 24]$$

$$= \frac{1}{12} (k+1)(k+2)(3k+8)(k+3)$$

$$= \text{LHS}$$

Thus if it is true for $n=1$ it is true for $n=2$ & hence $n=3$ etc.
 $\therefore$ it is true for all n ($n \in \mathbb{N}$)

$$c(i) f(x) = g(x) - \ln[g(x)+1]$$

$$f'(x) = g'(x) - \frac{1}{g(x)+1} \cdot g'(x)$$

$$= \frac{g'(x)[g(x)+1] - g'(x)}{g(x)+1}$$

$$= \frac{g(x)g'(x)}{g(x)+1}$$

$$(ii) \int \frac{\sin x \cos x}{\sin x + 1} dx$$

$$= \left[\sin x - \ln(\sin x + 1) \right]_0^{\frac{\pi}{2}}$$

$$= 1 - \ln 2$$

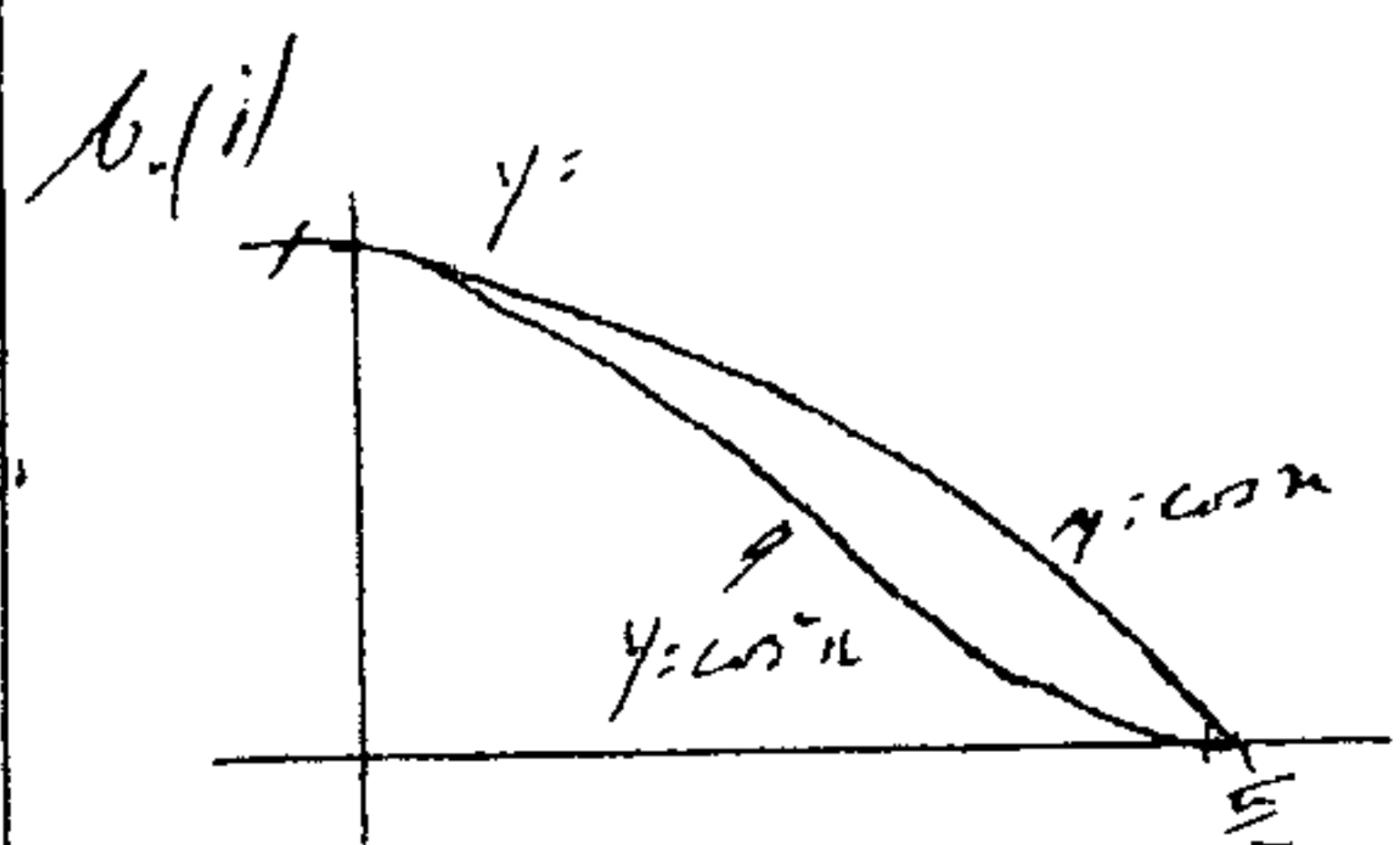
$$2 \cos^2 x = 1 + \cos 2x$$

$$4 \cos^4 x = 1 + 2 \cos 2x + \cos^2 2x$$

$$= 1 + 2 \cos 2x + \frac{\cos 4x + 1}{2}$$

$$8 \cos^4 x = 2 + 4 \cos 2x + \cos 4x + 1$$

$$= 3 + 4 \cos 2x + \cos 4x$$



$$ii \quad A = \int_0^{\frac{\pi}{2}} (\cos x - \cos^2 x) dx$$

$$= \int_0^{\frac{\pi}{2}} \cos x - \frac{1}{2}(1 + \cos 2x) dx$$

$$= \left[\sin x - \frac{\sin 2x}{2} - \frac{x}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \left(1 - \frac{\pi}{4}\right) \text{ u}$$

$$(iii) \quad V = \pi \int_0^{\frac{\pi}{2}} (\cos^2 x - \cos^4 x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \frac{1 + \cos 2x}{2} dx - \left(\frac{3 + 4 \cos 2x + \cos 4x}{8} \right)$$

$$= \pi \left[\left(\frac{x + \frac{\sin 2x}{2}}{2} \right) - \left(\frac{3x + 2 \sin 2x + \frac{\sin 4x}{4}}{8} \right) \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\frac{\pi}{4} - \frac{3\pi}{16} \right]$$

$$= \frac{\pi^2}{16} \text{ u}^3$$

a (i)

$$\cot x + \tan x = \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\sin x \cos x}$$

$$= \frac{2}{\sin 2x}$$

$$= 2 \operatorname{cosec} 2x$$

(ii)

$$\int_{\pi/6}^{\pi/3} 2 \operatorname{cosec} 2x \, dx$$

$$= \int_{\pi/6}^{\pi/3} [\cot x + \tan x] \, dx$$

$$= [\ln \sin x - \ln \cos x]_{\pi/6}^{\pi/3}$$

$$= \ln \tan x$$

$$= \ln \sqrt{3} - \ln \frac{1}{\sqrt{3}} = \ln 3$$

b. $a = b = (ab)^{1/2}$

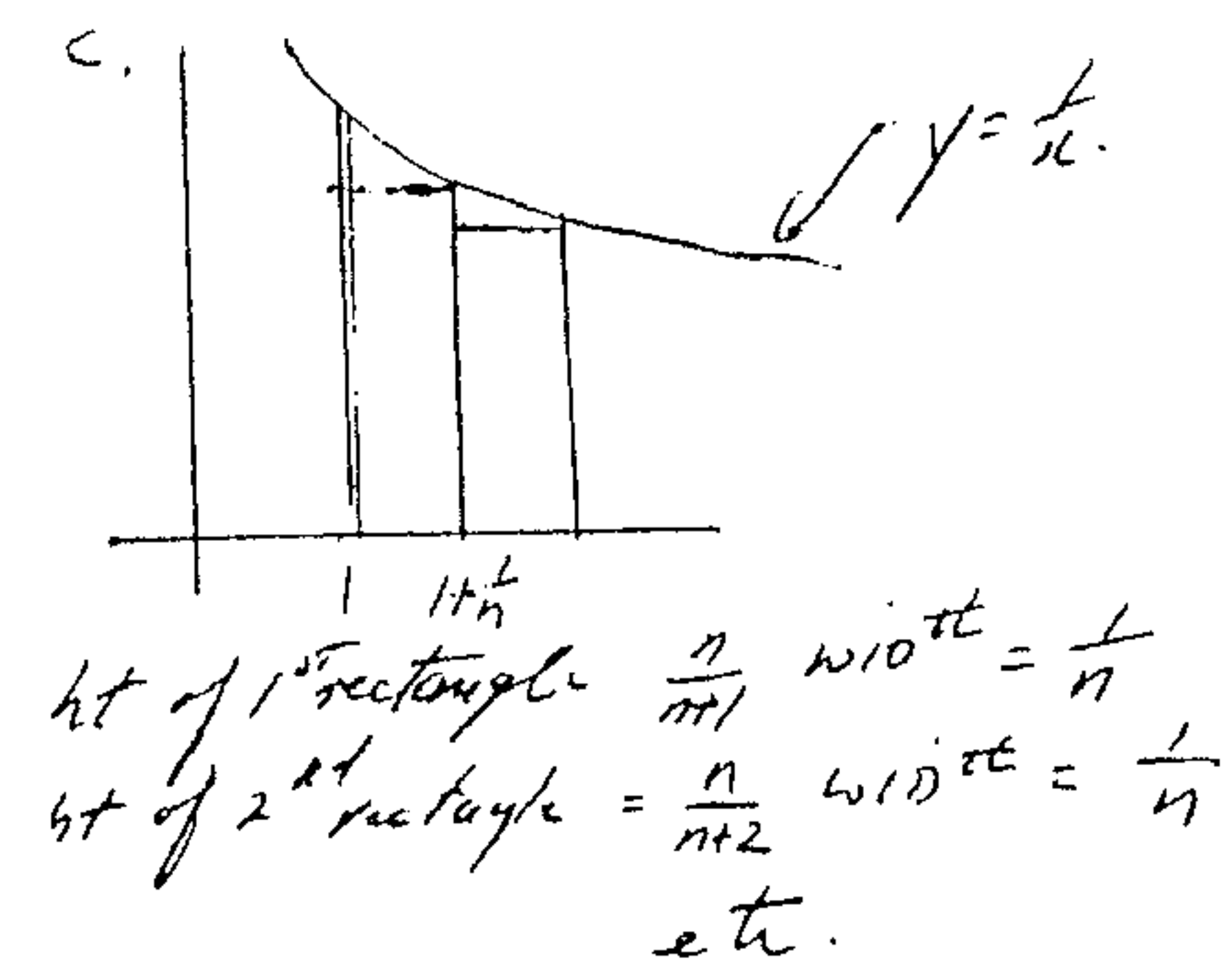
$$x \ln a = y \ln b = 3(\ln a + \ln b)$$

$$\frac{1}{x} + \frac{1}{y} = \frac{\ln a}{y \ln b} + \frac{1}{y}$$

$$= \frac{\ln a + \ln b}{y \ln b}$$

$$= \frac{\ln a + \ln b}{3(\ln a + \ln b)}$$

$$= \frac{1}{3}$$



Summing lower rectangles

$$\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$$

Taking the limit

$$A = \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right)$$

By integration $A = \int_1^2 \frac{1}{x} \, dx$

$$= [\ln x]_1^2$$

$$= \ln 2$$

2

$$\ln 2 = \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right)$$

3

b. $y = e^{kx}$

$$\frac{dy}{dx} = k e^{kx}$$

$$\frac{d^2y}{dx^2} = k^2 e^{kx}$$

Now $\frac{d^2y}{dx^2} + 7 \frac{dy}{dx} + 12y = 0$

$$k^2 e^{kx} + 7k e^{kx} + 12e^{kx} = 0$$

$$e^{kx} (k^2 + 7k + 12) = 0$$

$$\therefore k = -4, -3$$

6.6

Let F_n be amount in fund after n prizes are awarded.

(i)

$$\therefore F_1 = 2000 \times 1.025^2 - 150$$

$$= 81951.25$$

(ii)

$$F_6 = 2000(1.025)^{12} - 150(1 + 1.025^2 + 1.025^4 + \dots + 1.025^{10})$$

$$= 2000 \times 1.025^{12} - 150 \left(\frac{1.025^{12} - 1}{1.025^2 - 1} \right)$$

$$= 2689.78 - 1021.89$$

$$= \$1667.89$$

96 c.

$$i = 0 = 2000, 1.025^{2n} - 150 \left(\frac{1.025^{2n} - 1}{0.050625} \right)$$

$$2000 \times 1.025^{2n} - 2962.9629(1.025)^{2n} + 2962.96 = 0$$

$$1.025^{2n} = 3.0769$$

$$2n \ln 1.025 = \ln 3.0769$$

$$2n = 45.5$$

$$n = 22 \text{ prizes}$$